Microsoft 365 Copilot Analytics

Data Onboarding Guide

Upload methods, attribute mapping, verification, and maintenance — the complete org data playbook

Document Property	Value
Section	Setup
Audience	IT Admin, Insights Admin, Data Engineer
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How This Document Fits

This is a **Setup** guide. It is the complete step-by-step guide for uploading organisational data into Microsoft 365, covering all six upload methods, attribute mapping, verification, and ongoing maintenance. Read the Data Strategy first to decide *what* to upload and *why*, then follow this guide to execute.

Section	Purpose	Who Reads
Strategy	Decide what to implement	CIO, SA, Business Leaders
Setup (you are here)	Configure and deploy	IT Admin, Insights Admin
Operate	Use dashboards and reports daily	Leaders, Analysts, Managers
Reference	Look up details and fix issues	Everyone (as needed)

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1. Which Path Am I On? (Migration Diagnostic)

Microsoft is migrating all organisational data management from the Viva Insights web app to the M365 Admin Center. Use this quick check to determine which path applies to your tenant:

1. Go to `admin.microsoft.com` → **Setup** → **Migration and imports**
2. See **"Organizational Data in Microsoft 365"**? → You're **migrated** → follow Section 3
3. Don't see it? → You're **pre-migration** → follow Section 4
4. Unsure? → Check **Message Center** in the Admin Center for migration notifications

> **IMPORTANT: MODIS Migration — All Tenants Moving by Mid-2026**
>
> The **Microsoft Organizational Data in Microsoft 365 (MODIS)** migration is moving all tenants to the M365 Admin Center for organisational data management by mid-2026. Key changes:
>
> - **New role:** Organizational Data Source Administrator (replaces Insights Administrator for data uploads)
> - **Broader platform:** Org data now serves multiple M365 apps (Viva Insights, Copilot Dashboard, Microsoft 365 Profile, and more) — not just Viva Insights
> - **New capabilities:** HRIS connectors (SAP SuccessFactors, Workday), Azure Blob Storage, and API-based import — in addition to CSV upload
> - **Attribute naming:** Legacy names (`PersonId`, `ManagerId`) are replaced by `Microsoft_*` names (`Microsoft_PersonEmail`, `Microsoft_ManagerEmail`)
>
> If your tenant is still using the Viva Insights upload path, plan now to transition to the M365 Admin Center path.

2. Upload Methods Overview

Option	Method	Platform	Status	Refresh Behaviour
A	CSV Upload	M365 Admin Center	GA (primary path)	Incremental merge
B	CSV Upload	Viva Insights	Deprecated (mid-2026)	Full replacement
C	SAP SuccessFactors Connector	M365 Admin Center	GA	Auto-sync (weekly/mo
D	Workday Connector	M365 Admin Center	GA (≤100K users)	Auto-sync (JSON map
E	Azure Blob Storage	M365 Admin Center	GA	Automated CSV drop
F	API-Based Import	M365 Admin Center	Preview	Incremental refresh (R

NOTE: Decision Logic

- **Running SAP SuccessFactors?** → Use **Option C** (connector). Zero CSV.
- **Running Workday (≤100K users)?** → Use **Option D** (connector). Zero CSV.
- **Running Workday (>100K) or have a data warehouse?** → Use **Option E** (Azure Blob). Automated CSV pipeline.
- **Need programmatic control or custom HRIS?** → Use **Option F** (API). Incremental refresh.
- **Everything else?** → Use **Option A** (CSV to M365 Admin Center). Universal fallback.
- **Tenant not migrated yet?** → Use **Option B** (CSV to Viva Insights) temporarily.

TIP: Bypass On-Premises AD

If your HRIS data flows through **SAP SuccessFactors → middleware → on-premises AD → Entra ID**, be aware that Viva Insights only picks up 3 attributes from Entra (`PersonId`, `ManagerId`, `Organization`). Custom fields in Entra `extensionAttributes` are **not synced**.

Solution: Use the HRIS connector to send data directly to M365 — bypassing AD entirely for analytics purposes.

3. Upload Steps — M365 Admin Center (Primary Path)

This is the recommended path for all tenants that have completed the MODIS migration.

Step 1: Ensure Role Assignment

The person performing the upload must have the **Organizational Data Source Administrator** role assigned in Entra ID. This role is specific to the MODIS platform and is separate from the Insights Administrator role.

- Assign via: `admin.microsoft.com` → Users → Active Users → select user → Manage roles → search for "Organizational Data Source Administrator"

Step 2: Prepare Your CSV File

- **Encoding:** UTF-8 (with or without BOM)
- **File size:** <1 GB. For files >25 MB, upload via SharePoint first
- **Attribute naming:** Use `Microsoft_*` naming convention (e.g., `Microsoft_PersonEmail`, `Microsoft_ManagerEmail`, `Microsoft_Organization`)
- **No `EffectiveDate` column:** The M365 Admin Center uses a date picker in the UI, not a CSV column
- **No spaces in header names:** Use underscores (e.g., `Cost_Center` not `Cost Center`)
- **One row per employee:** Include every employee you want reflected in analytics

Step 3: Navigate to Data Connections

`admin.microsoft.com` → **Setup** → **Migration and imports** → **Organizational Data in Microsoft 365** → **Data Connections**

Step 4: Start Import

- **First upload:** Select **Get started**
- **Subsequent uploads:** Select **New import**

Step 5: Select Target Apps

Choose which M365 applications will receive the uploaded data:

- **Viva Insights** — required for Copilot Dashboard and Advanced Analytics
- **Copilot Dashboard** — enables org-level filters on the dashboard

- **Microsoft 365 Profile** — populates org data in user profiles

You can select all that apply. Each app may have its own required attributes.

Step 6: Upload File and Map Reserved Attributes

- Upload your CSV file
- Map the reserved attributes: `Microsoft_PersonEmail`, `Microsoft_ManagerEmail`, `Microsoft_Organization`
- The system will auto-detect columns that match `Microsoft_*` naming

Step 7: Review Custom Attributes and Configure Access

For each custom attribute, set the access level:

Access Level	Who Can See	Use Case
Public	All selected apps and their users	Department, Location, Job Function
Confidential	Only authorised analysts (Viva Insights only)	HourlyRate, Salary Band, Performance Rating
Custom	Specific apps only	Attributes shared with Profile but not Insights

Step 8: Validate and Submit

- Review the summary and submit
- **Processing time:** A few hours for small files, up to **3 days** for large files (>100K rows)
- Monitor status in the Data Connections page: **Validating** → **Processing** → **Complete**

4. Upload Steps — Viva Insights (Legacy Path)

> **DEPRECATED: This Path Is Being Retired by Mid-2026**
>
> The Viva Insights upload path is being replaced by the M365 Admin Center path (Section 3). Use this section only if your tenant has not yet been migrated. Plan to transition as soon as the migration completes.

Step 1: Prepare Your CSV File

- **Encoding:** UTF-8 (with or without BOM)
- **File size:** <1 GB
- **Include `EffectiveDate` column:** The date must be **earlier** than the upload date (e.g., if uploading today, use yesterday's date or earlier)
- Use legacy attribute names: `PersonId`, `ManagerId`, `Organization`

Step 2: Navigate to Data Hub

`insights.cloud.microsoft` → **Admin** → **Data hub** → **.csv upload** → **Start**

Step 3: Upload and Map Fields

- Upload your CSV file
- Map required fields: `PersonId`, `ManagerId`, `Organization`, `EffectiveDate`
- Map optional fields as needed

Step 4: Validate and Submit

- Review validation results
- **Processing time:** Up to **7 days** for large files
- Monitor status in the Data hub page

5. Attribute Mapping

This table maps legacy Viva Insights attribute names to the new M365 Admin Center naming convention. Use this when migrating from the legacy upload path or preparing a new CSV.

Legacy Name (Viva Insights)	M365 Admin Center Name	Required?	N
`PersonId`	`Microsoft_PersonEmail`	Required	E
`ManagerId`	`Microsoft_ManagerEmail`	Required	M
`Organization`	`Microsoft_Organization`	Required	D
`EffectiveDate`	N/A (date picker in UI)	Required (legacy) / Not needed (new)	L
`FunctionType`	`Microsoft_JobDiscipline`	Optional	D
`LevelDesignation`	`Microsoft_LevelDesignation`	Optional	E
`Location`	`Microsoft_CompanyOfficeLocation`	Optional	D
`HireDate`	`Microsoft_HireDate`	Optional	U
`HourlyRate`	`Microsoft_HourlyRate`	Optional	**C**
Custom (e.g., `BusinessUnit`)	Custom (same name)	Optional	U

HRIS Connector and Advanced Method Documentation

For detailed setup instructions on non-CSV methods, refer to Microsoft Learn:

- SAP SuccessFactors Connector
- Workday Connector
- Azure Blob Storage
- API-Based Import

6. Post-Upload Verification Checklist

After every upload, complete this checklist to confirm data quality before relying on dashboards.

Step 1: Check Import Status

Status	Meaning	Action
Success	All rows processed. No errors.	Proceed to
Partial Success	Most rows processed. Some rows had warnings (e.g., unmatched manager emails).	Download i
Failed	File rejected. No data was imported.	Check erro
Suspended	Processing was halted (usually by system or admin).	Check Data

Step 2: Download Import Log

Look for rows with unmatched `Microsoft_ManagerEmail` values. These indicate managers whose email addresses don't exist in the uploaded data or in Entra ID.

Step 3: Download Manager Log

This log shows employees whose `ManagerEmail` doesn't resolve to an active user in Entra ID. These employees will appear as "orphaned" in the manager hierarchy.

Step 4: Check Attribute Coverage

Navigate to the **Data Quality** tab. Every attribute should show **>95% coverage** (percentage of rows with non-blank values). Low coverage means dashboard filters will show incomplete results.

Step 5: Check Manager Tree Count

The number of root nodes in the manager hierarchy should be small (typically 1–5, representing CEO/top executives). If you see hundreds or thousands, many `ManagerEmail` values are missing or incorrect.

Step 6: Spot-Check Dashboard

- Open the Copilot Dashboard and verify the **Organisation filter** shows your uploaded values (not just Entra defaults)
- Confirm new employees appear in the population count
- Check that Job Function, Location, and other optional filters are populated (if uploaded)

Step 7: Approve Connector Data (If Applicable)

If using an HRIS connector (SAP SuccessFactors or Workday) with **"Manually approve"** enabled: you have a **48-hour review window** to inspect the data before it goes live. If not approved within 48 hours, the data is automatically approved.

7. Re-Upload & Maintenance

Refresh Behaviour by Method

Method	Refresh Behaviour	Key Differences
M365 Admin Center CSV	Incremental merge	Upload only changed or new rows. Existing rows r
Viva Insights CSV (legacy)	Full replacement	Must re-upload **all employees** every time. Missing
HRIS connectors	Automated (weekly/monthly)	No manual uploads required. Connector syncs on
API-based import	Incremental or full (configurable)	Controlled via automation. Supports programmati

Common Maintenance Scenarios (M365 Admin Center CSV)

Scenario	What to Upload
New hire	1 row with the new employee's data (incremental merge adds them)
Department change	1 row with updated `Microsoft_Organization` value (merge updates existing record)
Employee left	1 row with `Layer=-1` to remove the employee's organisational data
Historical correction	Use the **Historical import** option in Data Connections to submit data for a past date

Recommended Cadence

- **CSV upload:** Monthly (minimum). Set a recurring calendar reminder.
- **HRIS connectors:** Weekly or monthly (configured in connector settings)
- **API-based import:** As frequently as your automation runs (daily is possible)

WARNING: No Automatic Rollback

If bad data is uploaded, dashboard filters will break or show incorrect values until corrected data is processed (which may take up to 3 days). There is no “undo” button.

Pre-upload validation checklist:

1. Spot-check 10 random rows for accuracy
2. Verify every `Microsoft_PersonEmail` is non-blank and a valid UPN
3. Verify every `Microsoft_ManagerEmail` resolves to an existing user
4. Confirm no `TimeZone` column is present (system-generated; causes errors)
5. Confirm no spaces in header names (M365 Admin Center path)
6. Confirm no `EffectiveDate` column (M365 Admin Center path)

8. Data Freshness Monitoring

Where to Check

Check	Where	What to Look For
Upload / connector status	M365 Admin Center → Data Connections	Last import date, status (Success / Partial
Data freshness indicator	Viva Insights web app → Admin → Data hub	Freshness badge showing days since last
Attribute quality scores	M365 Admin Center → Data Quality tab	Coverage percentage per attribute (>95%
Dashboard spot-check	Copilot Dashboard → filters	Filter values match current org structure

Signs of Stale Data

Symptom	Likely Cause
New employees are missing from dashboards	Upload hasn't been refreshed since they were hired
Only one leader appears in Group view	Manager hierarchy data is outdated or incomplete
Old department/location values still showing in filters	Org changes not reflected in latest upload
Employee counts don't match HR system totals	Leavers not removed or new hires not added

Recommended Monitoring Cadence

When	What to Do
After every upload	Complete the Post-Upload Verification Checklist (Section 6)
Weekly (if using connectors)	Check connector health in Data Connections. Verify last sync date.
Monthly (if using CSV)	Confirm upload was completed. Spot-check dashboard filters.
Quarterly	Full audit: compare uploaded population against HR system of record. Review attribu

9. Key Rules

- **Upload for every employee** — include every person you want reflected in analytics, not just Copilot-licensed users
- **No `TimeZone` column** — TimeZone is system-generated from Outlook settings. Including it in your CSV causes a validation error.
- **M365 Admin Center path: no `EffectiveDate` column** — the date is selected via the UI date picker
- **M365 Admin Center path: no spaces in header names** — use underscores (`Cost_Center`, not `Cost Center`)
- **Viva Insights legacy path: include `EffectiveDate`** — must be a date earlier than the upload date
- **Incremental merge is the default** in the M365 Admin Center — you only need to upload changed or new rows. Unmentioned employees retain their existing data.
- **Data precedence: Entra ID takes precedence by default** — for attributes like Department, Entra values win unless you configure the override in Data Connections settings
- **One import at a time per tenant** — if an import is in progress, you must wait for it to complete before starting another

10. Common Upload Errors

Error	Cause
Blank `Microsoft_PersonEmail` rows	Rows with empty email addresses
Missing required attributes for selected apps	Target app (e.g., Viva Insights) requires specific attribu
Empty data rows	Rows with all-blank values (often trailing rows from Exc
Sending data to unlicensed apps	Selected a target app that your tenant doesn't have lic
Custom attributes sent to unsupported apps	Not all apps support custom attributes
Attribute name conflicts with `Microsoft_` reserved names	Custom attribute uses a name that starts with `Micros`

For a comprehensive list of validation error codes and resolutions, see Upload error troubleshooting on Microsoft Learn.

11. What to Read Next

Ready to...	Read this
Look up the full attribute reference	Attribute Reference (Reference layer)
Troubleshoot a failed upload	Troubleshooting Guide (Reference layer)
Revisit the data strategy decision	Data Strategy (Strategy layer)
Configure roles, VFAM, and dashboards	Admin Setup Guide (Setup layer)
Validate a customer's org data file before upload	Org Data Validation Prompt (AI-assisted validation tool)

Disclaimer

This guide is provided as a procedural reference. Always refer to Organizational Data in Microsoft 365 on Microsoft Learn for the most up-to-date information. Upload interfaces and attribute requirements may change as Microsoft completes the MODIS migration.